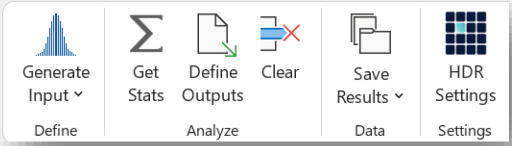

Probability Management

GUIDE TO CHANCECALC MONTE CARLO

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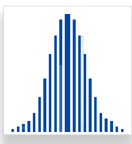
1. Generate Input



The famous mathematician and father of the modern computer, John von Neumann, said *“Anyone who attempts to generate random numbers by deterministic means is, of course, living in a state of sin.”* This did not mean that he thought it could not be done; in fact, his first computer launched the field of Monte Carlo simulation. However, generating such numbers is not straightforward and, some 70 years after the advent of Monte Carlo simulation, this is still an active research area. As described below, **Chance**Calc Monte Carlo provides internal random number generation, as well as the option to use Stochastic Libraries in CSV, SIPmath™ 2.0 and SIPmath 3.0 Standard.

The distributions currently supported are listed below.

1. Uniform
2. Beta
3. Binomial
4. Chi Square
5. Discrete
6. Exponential
7. F
8. Gamma
9. Lognormal (Percentile)
10. Lognormal (Mean and Standard Deviation)
11. SPT Metalog
12. Myerson
13. Normal

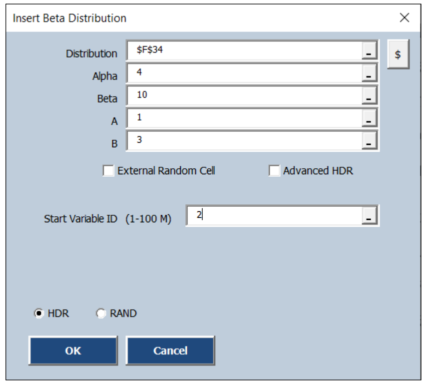
14. PERT
15. Poisson
16. Resample
17. Triangular
18. T
19. Weibull
20. Correlated Normal
21. Correlated Uniform

All the distribution functions in **Chance**Calc Monte Carlo are driven by Uniform random numbers between 0 and 1. These are referred to as U(0,1) variables. There are two options for U(0,1)s to drive your simulations.

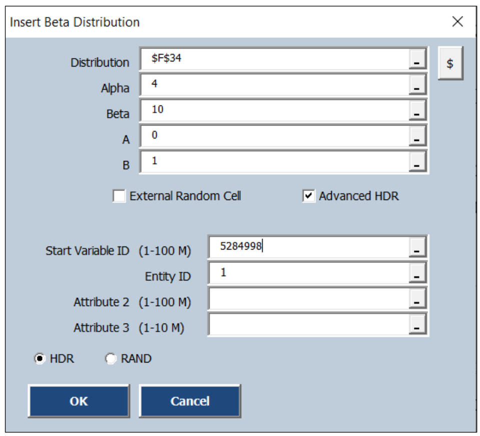
1) The Native Excel `RAND()` function.
   This computes results the fastest, but does not provide replicable results. We STRONGLY DISCOURAGE its use as it can greatly slow down other SIPmath models.
2) The HDR Generator
   A Random Access Pseudo-Random Number Generator from Hubbard Decision Research (the HDR Generator). This is an evolving family of simple generators that use pure Excel formulas to provide repeatable results in an interactive simulation environment. Each instance of an HDR U(0,1) must have a unique Variable ID to avoid correlated results. The version of the HDR used in the tools has passed some basic random number tests, but should not be considered to be a "pedigreed" generator. However, this is the suggested approach for convenient everyday modeling. Because the modular nature of SIPmath, you may always go back and replace your U(0,1)s at a later date.

1.1 SAMPLE DISTRIBUTION DIALOG:

Distribution: Enter the cell or range of cells where you want the formula for the chosen distribution to appear. A range must be either a single row or a single column.



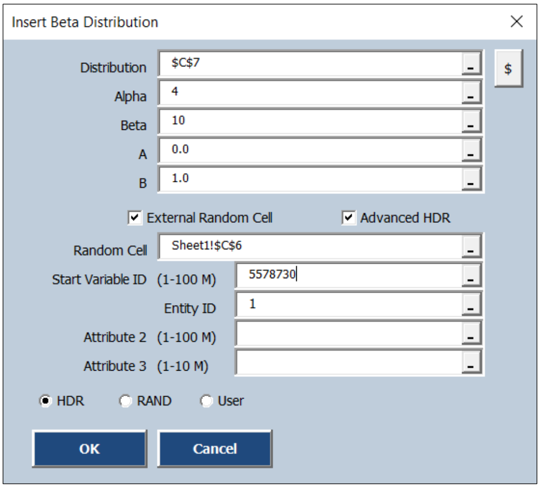
$: The $ button changes the address of an input parameter from absolute to relative. Click in the box of the parameter you want to change and then press $ to change the address. The $ button will cycle through the various combinations of holding the Row constant, Column constant or both Row and Column Constant. If you don't use the $ button, then **Chance**Calc will ignore $s and make a best guess whether to use absolute addresses when entering the distribution formula; usually **Chance**Calc will use relative addresses, unless the input parameter is a single cell and the distribution address is two or more cells.



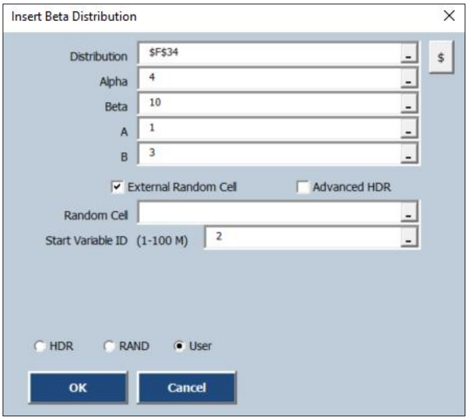
Alpha, **Beta**, **A, B**: These parameters determine the pattern of results computed by the formula for the Beta distribution. They may be entered either as constant values, single cells, or a range of cells. In the latter case, the range must be the same size and shape as the Distribution range. Note that every distribution has its own set of input values.

External Random Cell: check this box if you want to either correlate your distribution with other distributions or specify your own U(0,1)s to drive the Distribution formula. The External cell (or range of cells) will be used as random inputs to the formula. For instance, a Normal distribution in cell A1 might be computed as

`=NORMINV(A2, 0, 1)`, where `A2` would be the External Random Cell containing a random uniform.



Leave the box unchecked if you want the Distribution formula to use a random value computed inside the formula (RAND or HDR only). For instance, a Normal distribution in cell A1 might be computed as `=NORMINV(RAND(),0,1)`



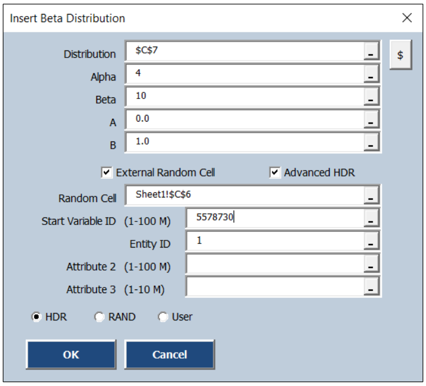
Checking the box shows the Random Cell field below; unchecking it will hide the Random Cell field.

Advanced HDR: Checking this box activates the three additional HDR parameters. One parameter is always named Entity ID. The other two may have their name changed in the HDR Settings dialog. The parameters can be used as additional "seed" values to avoid seed collisions. For example, Entity ID could be used to distinguish departments within an organization.

Random Cell: If External Random Cell is checked, enter a cell or range of cells that will contain the random values. If this is a range, it must have the same size and shape as the Distribution range.

HDR: Select this option to generate pseudo-random values with the Hubbard Decision Research generator. This causes the sequence of pseudo-random values to be repeated with the same inputs every time a model is calculated, allowing for repeatable and auditable results. External Random Cells will be filled with the HDR formula for generating pseudo-random values.

Start Variable ID: This is required when using the HDR generator. Enter a constant, a cell, or a range of cells that has a value to identify the variable ID for the HDR generator. If you use a range, it must have the same size and shape as the Distribution range. This input can also be thought of as the "seed" for the random distribution. Having a unique Start Variable ID or seed is important to ensure the independence of the random variables.



Entity ID: This is an additional "seed" value for the HDR generator. The value can be changed with the HDR Settings dialog. Ordinarily it should be the same for all users in each group (or "Entity") within an organization.

Attribute 1 (& 2): These serve as additional seed values. The names can be changed in the HDR Settings dialog.

RAND: Select this option to generate pseudo-random values with Excel's `RAND()` function. External Random Cells will be filled with `=RAND()` formulas.

User: Select this option to use your own source of external random values. If this option is chosen, the Random Cells specified will be left unchanged, so you can put whatever formula you prefer in the Random Cells. For example, by using the Library Input command to specify data from a known generator.

OK: Press this button to fill the Distribution range with formulas to compute the chosen distribution.

Cancel: Press this button to close the dialog without doing anything.

The example above shows the dialog for the Beta distribution. The other dialogs, except for the Myerson, Poisson, Triangular, and Correlated distributions (marked by [1]) are very similar, differing only in that the input values **Alpha**, **Beta**, **A** and **B** will have different names, and not all dialogs have 4 inputs.

Distribution	Input parameters			
Uniform	<none>			
Beta	Alpha	Beta	A	B
Binomial	Number of Trials	Chance of Success		

[1] These distributions require an External Random cell to hold the U(0,1).

Distribution	Input parameters			
Chi-Square	Degrees of Freedom			
Discrete	Values	Probabilities	Cumulative Probabilities	
Exponential	Alpha			
F	Degrees Free 1	Degrees Free 2		
Gamma	Alpha	Beta		
Lognormal (percentile)	50th percentile	Top percentile rank	Top percentile value	
Lognormal (mean, std dev)	Mean of underlying normal	Std Dev of underlying normal		
SPT Metalog[1,2]	Low, 50th, and high percentiles	Low-percentile probability	Bounds (if any)	
Myerson [1]	10th percentile	50th percentile	90th percentile	
Normal	Mean	Standard Deviation		
PERT	Low value	Most-likely value	High value	
Poisson [1]	Lambda			
Resample	Data to Resample			
Triangular [1]	Minimum	Most Likely	Maximum	
T	Alpha			
Weibull	Lambda	K		
Correlated Normal [1]	Mean	Covariance Matrix		

Distribution	Input parameters			
Correlated Uniform [1]	Correlation Matrix			
1: These distributions require an external random cell(s) to hold the U(0,1) 2: These are SPT Metalogs. See below for more information.				

1.2 BRIEF DISTRIBUTION DESCRIPTIONS

The descriptions below are indications of where these distributions may be applied. We strongly recommend consulting appropriate published references before applying any of these in practice.[2]

Uniform: Every value between 0 and 1 is equally likely. Drives all other generators.

Beta: Used to model random variables limited to an interval. For example, the percentage of defective parts in a batch (between 0 and 100%).

Binomial: Gives the number of successes in the given Number of Trials with the given Chance of Success. It can give any whole number from 0 to ∞. Note that, in Excel, ∞ is about 10^{15}.

Chi-Square: The chi-square distribution with K degrees of freedom is the distribution of the sum of the squares of K independent standard normal random variables. It can give any value from –∞ to ∞.

Discrete: Gives a distribution of user-specified values which occur with user-specified probabilities. The Cumulative Probabilities field must be a

[2] These distributions require an External Random cell to hold the U(0,1).

blank range with the same size and shape as the Probabilities field; it will be filled in by **Chance**Calc with the correct formulas to compute this distribution.

Exponential: Gives the time between events, when the average time is alpha. Can give any value greater than or equal to 0. (For the distribution of the number of events in a given time period, see the Poisson distribution.)

F: Used to test whether two data sets have the same degree of diversity. If U and V are chi-square distributions with degrees of freedom given by `Degrees Free 1` and `Degrees Free 2`, then F `is (U/Degrees Free 1)/(V/Degrees Free 2)`. Can give any value from 0 to ∞.

Gamma: Gives the waiting time until alpha events have happened, when the average time between events is Beta. (Thus when Alpha=1, this is the same as the exponential distribution.) Alpha and Beta are (possibly fractional) values > 0.

Lognormal (percentile): The log of this distribution is a normal distribution. The top percentile rank can be between 0.6 and 0.99. If this rank is (say) 0.75, then 75% of the generated values will fall below the top percentile value. Can give any value from 0 to ∞.

Lognormal (mean & std dev): The log of this distribution is a normal distribution. The mean and standard deviation are those values for that normal distribution. I.e. the log of this distribution will have the given mean, and similarly for the standard deviation.

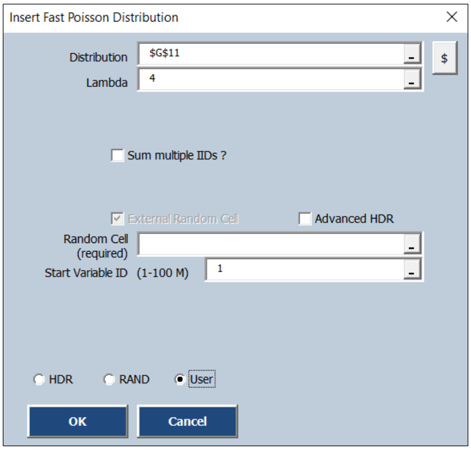
Myerson: Distribution, invented by Roger Myerson, is a generalization of both the Normal and Lognormal distributions, and can imitate either. If the 10^{th}, 50^{th} and 90^{th} percentiles are evenly spaced (e.g. 3, 5, 7) then the distribution is a Normal distribution with the specified percentiles. If the percentiles increase geometrically (e.g. 1, 4, 16) then the distribution is a Lognormal. Other patterns will give intermediates between Normal and Lognormal.

Normal: Also called the bell shaped curve, or Gaussian distribution. It has the mean and standard deviation specified by the inputs. Can give any value from -∞ to ∞.

PERT: Similar to a triangular distribution. The PERT distribution is a smooth curve rather than a triangle with a sharp peak. The minimum, most likely, and maximum values are specified in the inputs.

Poisson: The number of events (such as arrivals of customers) in a given time interval, given

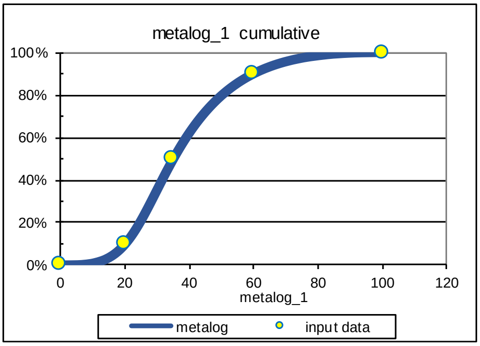
an average arrival rate, Lambda. (For the distribution of time between two events, see the Exponential distribution.)

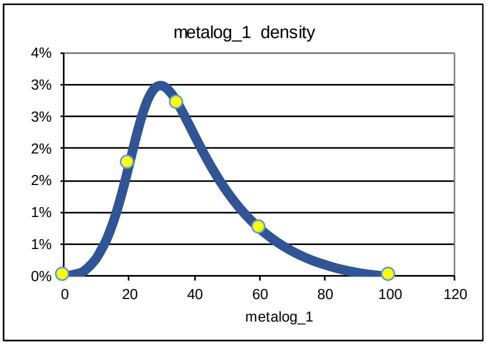


Resample: Chooses a random value (i.e. re-samples data with replacement) from a row or column of cells specified by the user.

SPT Metalog: Distribution family, invented by Tom Keelin, is comprised of unbounded, semi-bounded, and bounded distributions, each of which has nearly unlimited shape flexibility. All metalogs are quantile-parameterized, meaning that they are parameterized by (probability, quantile) points on the cumulative. Known lower and/or upper bounds are optional parameters. If no bounds are specified, the metalog is unbounded.

All metalogs have closed-form equations for both cumulative and density functions. The graphs below show a bounded metalog parameterized by cumulative-points (0.1, 20), (0.5,35), and (0.9,60), along with lower and upper bounds of 0 and 100 respectively.





To generate an SPT Metalog distribution, you should first fill in a block of data containing the names of the metalog distributions you are defining and the low, medium, and high quantile values for each distribution. Do this before invoking the *SPT Metalog* tool. You can arrange the distributions either stacked:

7	delta	4	5	6
8	echo	5	6	8
9	foxtrot	6	7	10

(Metalog "delta" is row 7, "echo" is row 8, and "foxtrot" is row 9. The low quanties are the first column of numbers, 4, 5 and 6. The median quantiles are the second column, 5, 6 and 7. The high quantiles are the third column, 6, 8 and 10).

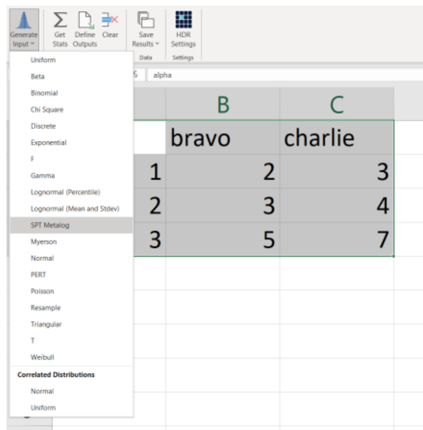
Or side-by-side:

	A	B	C
1	alpha	bravo	charlie
2	1	2	3
3	2	3	4
4	3	5	7

(Metalog “alpha” is column A, “bravo” column B, and “charlie” column C. Low quantiles in row 2, median in row 3, high in row 4.)

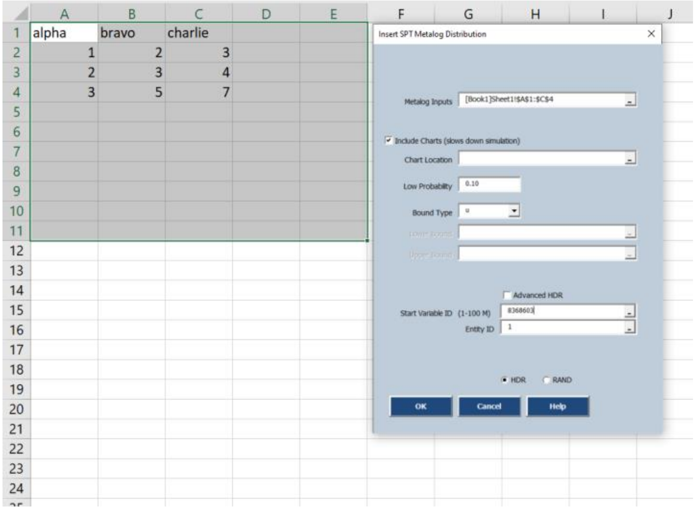
Once you have defined the metalogs and their quantiles, select the range containing them

	A	B	C	D
1	alpha	bravo	charlie	
2	1	2	3	
3	2	3	4	
4	3	5	7	
5				



and invoke the SPT Metalog tool.

ChanceCalc will increase your selection by a few rows and columns. This shows where **Chance**Calc will add additional information when creating the SPT Metalog distributions. The dialog box allows you to specify the following values: Where the PDF and CDF charts (if any) will be placed; the probability that will be used for the low quantiles of the SPT distributions (Low Probability); whether the distribution is bounded above, below, both, or neither (Bound Type); the lower and upper bounds (if any); and parameters for the HDR random value that will be used to create the SPT distribution.



Although **Chance**Calc provides the HDR generate as part of the SIP distribution, you may find it more convenient to add your own random values after creating the SIPt distributions.

The metalog implementation in **Chance**Calc currently encompasses the SPT (symmetric percentile triplet, e.g. 10^{th}, 50^{th}, and 90^{th} percentile) metalogs. For a tutorial on using SPT metalogs in the SIPmath Tools, see http://www.metalogdistributions.com/downloads/sipmath.html. For more general information, including on SPT metalogs, on general metalogs (even more flexible metalogs parameterized by an unlimited number of data points), equations, publications, and downloadable Excel workbooks, see [www.metalogdistributions.com.](http://www.metalogdistributions.com)

Triangular: The minimum, most likely, and maximum values are as specified in the inputs.

T: Also called Student's t-distribution. Similar to a Normal distribution, but more likely to have extreme values. The larger Alpha is, the closer this is to a Normal distribution.

Weibull: This can be interpreted as the time-to-failure of a process. If $k > 1$, the rate of failure increases with time. If $k < 1$, it decreases. K must be > 0. Lambda characterizes the average time to failure; about 63.2% of the values will be less than Lambda.

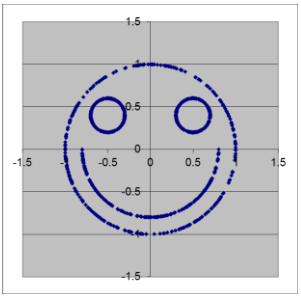
Correlated distributions: Creates sets of distributions which are correlated as specified by the user. Either uniform or normal distributions can be correlated. More detail below.

1.3 DISTRIBUTIONS REQUIRING EXTERNAL RANDOM CELLS

The Myerson, Poisson, Triangular, Correlated Uniform and Correlated Normal distribution formulas require the use of a Random Cell. Those dialogs have no External Random Cell checkbox, merely an External Random Cell field.

1.4 CORRELATED DISTRIBUTIONS

It is often desirable to generate interrelated, or statistically dependent random variables. In classical statistics this was generally described in terms of ***correlation***, which measures the degree to which the scatter plot of two variables lie along a straight line. Like many concepts from classical statistics, correlation is most applicable to normal distributions. But for many sorts of variables correlation is misleading. For example, the X and Y values of the scatter plot on the right have a correlation of zero, yet there is an obvious interrelationship. A major benefit of SLURPs is that they preserve such patterns within the data, so the interrelationships are handled automatically. This captures, a much broader set of relationships than correlation, and may be accomplished through the Resample distribution.

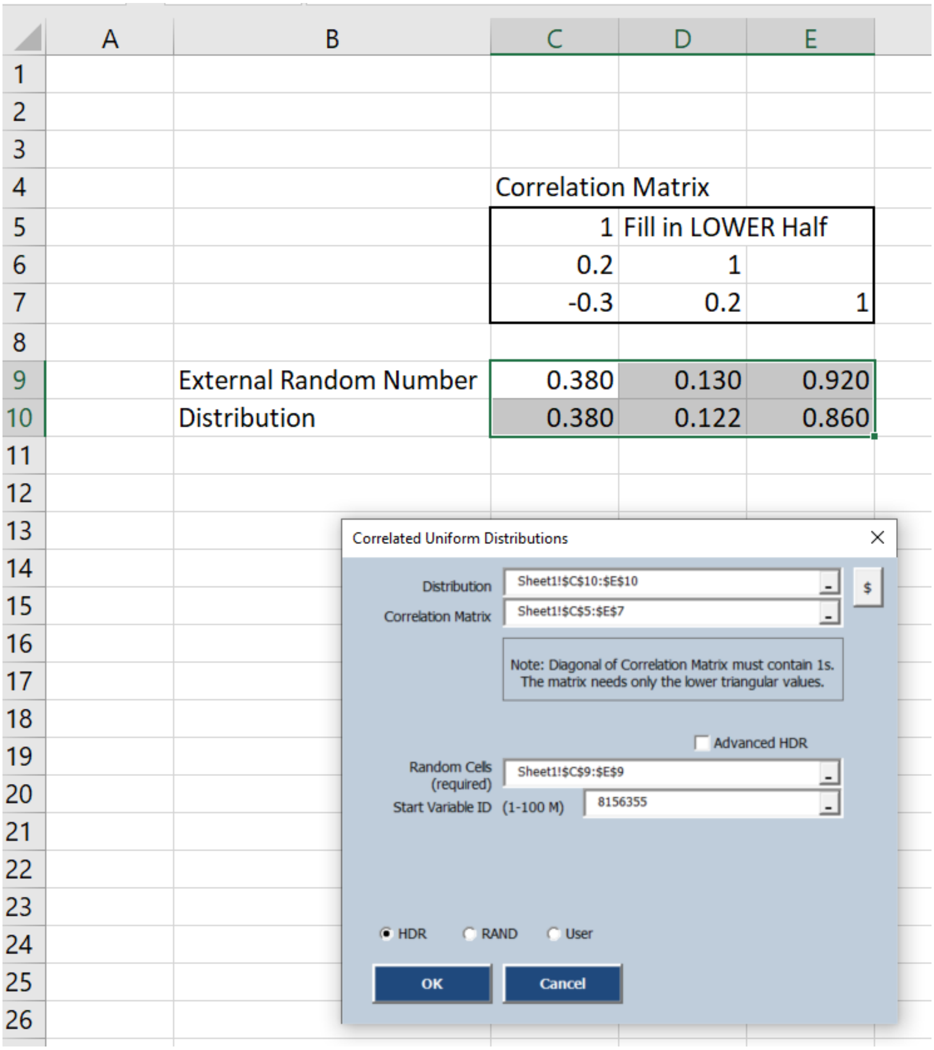


For those trained in statistics, and familiar with Pearson's and Spearman's definitions of correlation, the Tools provide two types of correlated distributions. These generate sets of Uniform or Normal pseudo-random values which have user-specified correlations. When these distributions are created, **Chance**Calc adds a new worksheet to your model named SIPmath Cholesky Decomposition. This sheet contains intermediate formulas that are required to compute the correlated values.

1.4.1 Correlated Uniform Distribution

The primary reason to create correlated uniforms is to drive other distributions through external random cells. For example, one could generate Poisson and Triangular variables that were correlated to one another.

Distribution: Enter the range of cells (a row or column) where you want the correlated random values to appear.



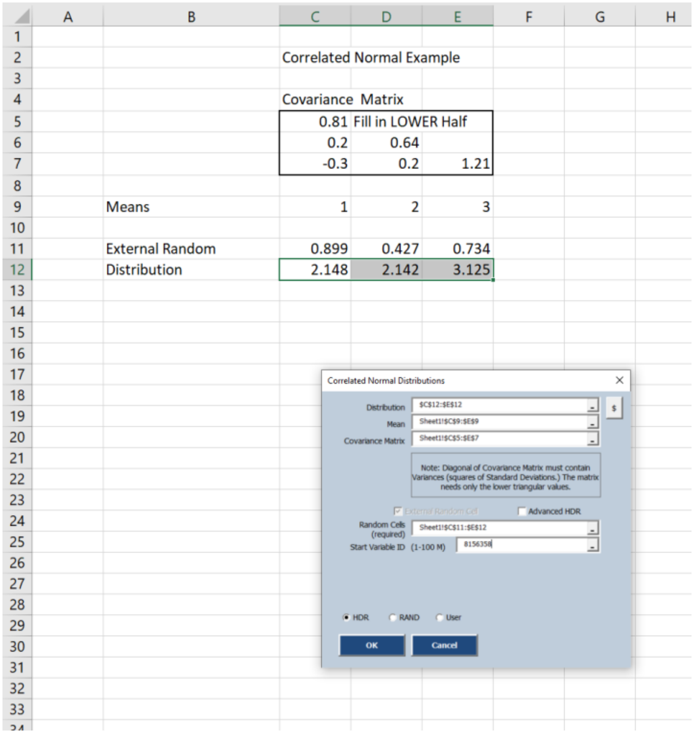
Correlation Matrix is a square range of cells containing the correlations. The main diagonal of this matrix must contain the value 1. The correlations should be put below the main diagonal; values above the diagonal will be ignored. The range must have as many cells in its sides as there are cells in the Distribution range. For instance, if the Distribution range is four cells, the Correlation matrix must be a 4x4 range.

NOTE: The correlation entered for the uniform is approximately the Spearman's Rank correlation, which is quite close to the Pearson's correlation.

Random Cells is a required range that is parallel to the Distribution range.

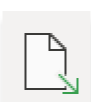
1.4.2 Correlated Normal Distribution

Mean: Enter the mean value of the normal distributions here, either as a single constant, or as a cell or cell range. If this is a range, it must have the same size and shape as the Distribution range.



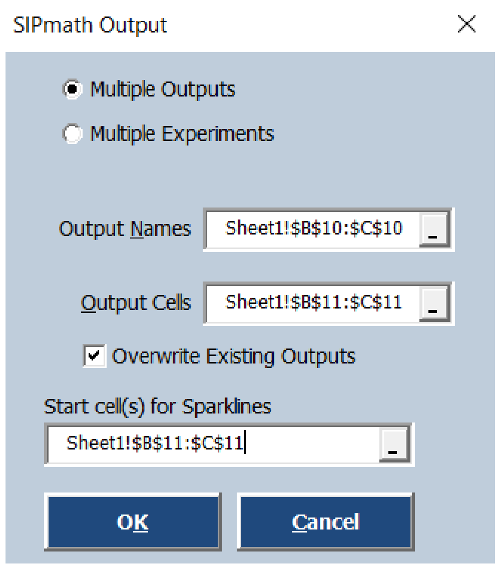
Covariance Matrix is like the correlation matrix for correlated uniform distributions, except you use covariances and variance instead of correlations. Note, the variances are the squares of the standard deviations of the normals.

2. Define Outputs



Define Outputs is actually two different dialogs: Multiple Outputs, which is displayed in the examples, or Multiple Experiments, for "What If", analysis. The dialogs work the same way in both SIP Library and Generate Mode, but the Experiments setting is less informative in Generate Mode.

Warning: If Multiple Experiments is used with an existing Multiple Outputs model, the current PMTable worksheet is overwritten by the analysis output. Thus, any model calculation or graph that depends on the earlier PMTable output will have errors. You should first use the Save PMTable feature to save the Multiple Outputs DataTable and generate a new PMTable. See description above.



2.1 MULTIPLE OUTPUTS

Multiple Outputs runs a single set of trials through one or more output cells, each of which will correspond to a column in the PMTable sheet. This dialog lets

you specify names for (a) the simulation result distributions and (b) the cells containing the formulas to be simulated.

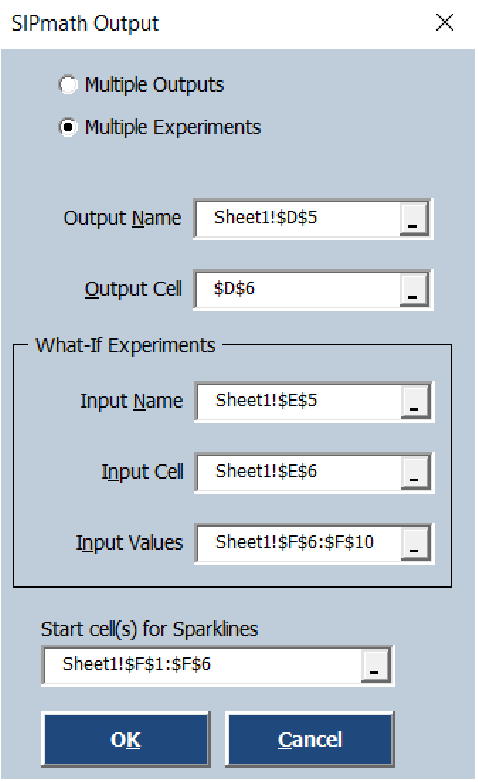
If you specify a Start Cell(s) for Sparklines range in the dialog, then the cells in that range will have a light blue sparkline histogram and a black border.

Define Outputs will, for each output cell, put a named column range in PMTable, using the associated name that you specify. These names, referring to the output distributions, can be used when creating graphs and summary statistics, using native Excel formulas anywhere in your model.

You may specify ranges of output cells. If you check "Overwrite Existing Outputs," a new table will be created, overwriting any data currently in place. Otherwise it will append your selections to the right of the current output columns. This means you don't have to specify all your outputs in one shot. Also, the output cells don't have to be contiguous.

2.2 MULTIPLE EXPERIMENTS:

You get the second dialog by selecting **Multiple Experiments**. The purpose of this tool is to find out how changes in one particular value assumption impact the model results. The formal name for this is sensitivity analysis.



In this context, an 'experiment' is a single run of the model, and the 'What If' analysis consists of some number of experiments, driven by an input variable that changes value for each experiment.

The 'Input Values' field holds the list of values that you will use to drive the analysis. The model will be run once for each of these values. 'Input Name' and 'Input Cell' address the cell that you will use to pass successive values from the Input Values list to the model.

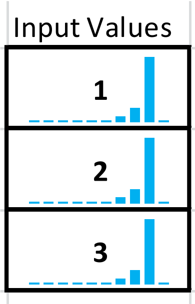
For each model run, the 'Output Cell' trial values will be appended to PMTable, using a naming convention that prevents collisions and makes it easy to refer to analysis results for populating reports and graphs. Each PMTable column is given a range name taken from the names assigned in the **Chance**Calc Output dialog. That is:

`OutputName_InputName_InputValue`.

For example, if the output name was “Cost”, the input name was “Stock”, and the input stocking levels were 10,15 and 20, then the column names in the PMTable sheet would be: Cost_Stock_10, Cost_Stock_15, and Cost_Stock_20.

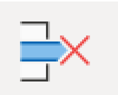
Setting up the analysis is the same as setting up for any model, except that you can define only one output, and you need to define the input list to drive the analysis.

Note that the experiment Input Values are the values you want to test, not the same as the input distributions you specified in the Define Inputs dialog, which will still drive each model run. The “What-If” values will drive the multiple experiments.



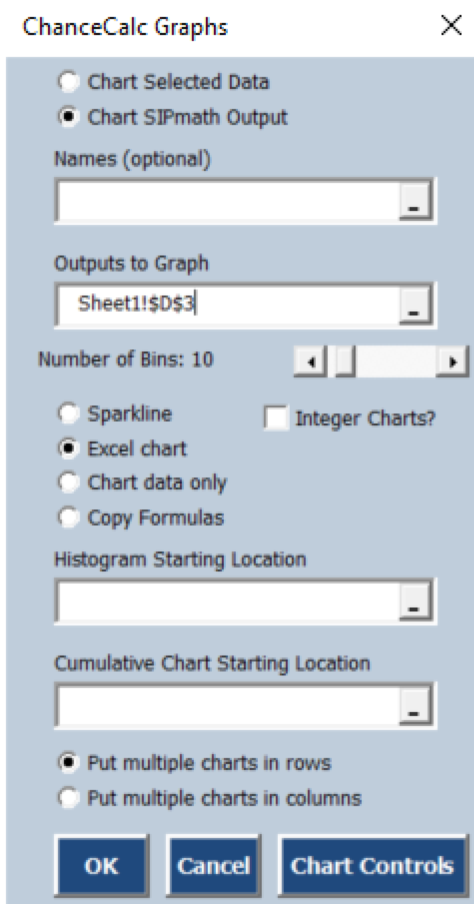
If you specify a Start Cell(s) for Sparklines range in the dialog, then the cells in that range will have a light blue sparkline histogram and a black border.

3 Clear

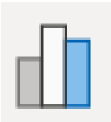


Clicking the *Clear* tool once removes input and output formulas and formatting from the selected cells. It also removes any related chart data from the **Chance**Calc Chart Data worksheet. With an output variable selected, clicking the clear button once will remove the output variable from the PMTable worksheet. This will reduce the size of the model and speed up the calculation time. But graphs can no longer be generated. Clicking the Clear tool a second time will delete the formula. If the cell selected is a random variable but not an output, then the cell contents are deleted. This can also be accomplished by using the keyboard delete key.

Clear also provides a simple way to remove the relatively slowly-recalculating formulas that compute data for the **Chance**Calc cumulative probability (CDF) chart. If a **Chance**Calc CDF chart for an output is selected when Clear is pressed, the chart and the formulas for computing the cumulative chart will be removed. Histogram charts and their formulas will be left alone.



4 Graphs



The *Graphs* button lets you create histograms and cumulative graphs from either ***Chance**Calc Output Cell(s)* or pre-existing *Selected Data* ranges.

Specify a location to put the Histogram(s) and/or a separate location for the cumulative. If only the Histogram field has a cell reference, then the data for the CDF will not be generated. If the histogram cell reference is not present, then the histogram graph will not be generated but the data will be calculated on **Chance**Calc Chart Data. The only way to not generate the histogram data is to not request Sparklines when defining the output variable and not specifying either a histogram and CDF graph.

For pre-existing data, select a range of values to be charted. You may either: 1) select that range before pressing the *Graphs* button, 2) type the range specification or name of the set in the Data field manually, or 3) click in the Data field and select the range to be charted.

The Number of Bins slider sets the number of histogram bars. (Cumulative (or CDF) charts are always charted with 105 points using Excel's `PERCENTILE()` function.)

If you check the Integer Charts checkbox, *Graphs* will assume that the data to be charted consists of integers (whole numbers only), such as 1, 2, 87, and −452. The number of bins will be automatically determined by *Graphs*, and the slider control will be greyed out when you check Integer Charts. The histogram will have one bar for each whole number between the smallest and largest whole number in the data, up to 100 bars. The bars will be labeled as whole numbers.

Note This option is not recommended if the range of data from smallest to largest value is more than 100.

If you select Sparkline[3] or Excel Chart the dialog will expand to ask you which graphs you want and where you'd like them put. Note that sparklines are not available for CDF charts.

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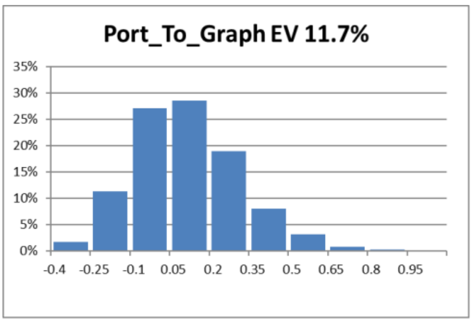
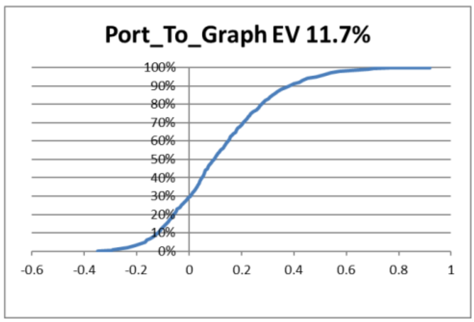
[3]In Excel 2010 or later.

If you are charting more than one set of data, additional charts can be placed in a row to the right of the first chart, or in a column below the first chart. You can select which with the radio buttons at the bottom.

Pressing OK will create the chart data and Sparkline or Excel chart, if any, in the specified location. Here's what histogram and cumulative distributions might look like once created:





The the series label is the output name followed by the expected value of the distribution. On the **Chance**Calc Chart Data there is an option to change the CDF to an Exceedance chart. An Exceedance chart plots the probability of being a certain value or more, whereas the CDF graph plots the probability of being that value or less.

CDF charts use Excel's `PERCENTILE()` function, which can be slow when used with a large number of trials. If you find the CDF chart is slowing down Excel's response time, you can remove the CDF chart and the `PERCENTILE()` functions by selecting the CDF chart and pressing the **Chance**Calc *Clear* button.

4.1 **CHANCE**CALC CHART DATA

	A	B
1		hdr
2	Expected Value	0.5
3	Series Name	hdr Avg 0.
4	Integer chart?	FALSE
5	Loss Exceedance	FALSE
6	Number of Bins	10
7	Label position	10
8	Bin Width	0.1
9	Decimals	2
10	Scale	1
11	Format Char	General
12	Use Axis Title?	FALSE
13	Axis Title	Axis Title
14	Min	0
15	Bin Range	0.1
16	Frequency	0.2
17	Labels	0.3
18	Cumulative	0.4
19		0.5
20		0.6

When you initialize a model or click Ok to close the *Graphs* dialogue box under the Select Data option, a new worksheet will appear titled "**Chance**Calc Chart Data." Here is a screenshot of example chart data.

The **Chance**Calc Chart Data will store information on every Output Variable. Each variable will occupy one column. The rows contain different types of data. The color of the data indicates the type of data. Row 1 will have the Output Variable name and Row 3 will specify the Series Name that appears in either the histogram or CDF graph.

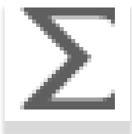
The Format Char row allows you to specify the formatting that will be used for labels on the CDF and PDF charts. Clicking on the Format Char cell activates a drop-down control with a list of the allowed formats. (The formats are listed in the red cells A36:A48 on the **Chance**Calc Chart Data sheet. The formats are defined in the red cells A22:A34.) If you choose a Custom format, then enter the custom format string in cell A34.

The worksheet uses the outline function to facilitate displaying only information desired. For example, to see only the CDF data, click on the "1" outline and then expand the last set of data for the Cumulative. When a Sparkline graph, histogram or CDF graph is requested the sheet is set back to outline “2”.

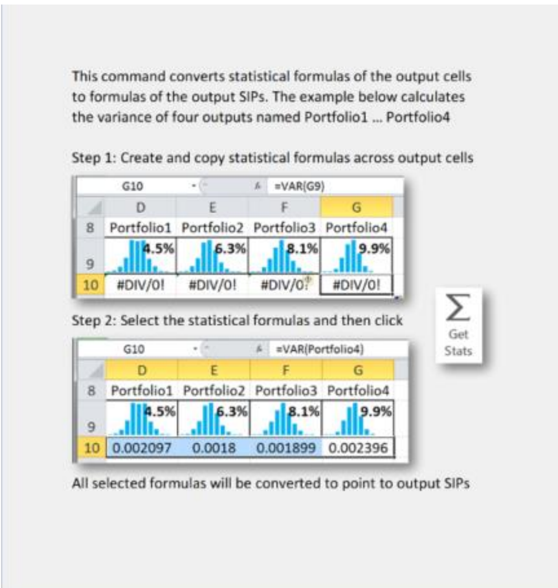
Data in this worksheet can be manually edited to change the way your output histogram on the other sheet looks. Bins can be manually edited, an offset can be applied, the number of decimal places allowed for class widths can be changed, and the histogram bar sizes (frequencies) can be altered.

Selecting “Chart Data Only” in the *Graphs* box will make the chart data table without creating the chart. You can then use this to create your own charts using Excel’s *Insert Chart* tool. The ranges for the bins are suggestively named (e.g. Profit_bins, Profit_freq, assuming your data had a heading of “Profit”).

5 Get Stats

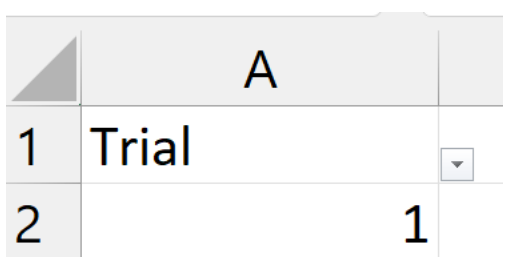


The *Get Stats* tool is designed to simplify the application of statistical formulas such as `VAR()`, `PERCENTILE()`, etc. to model outputs. Output variables are essentially SIPs (arrays of observed realizations), and must be referred to by the range names that were automatically created when they were defined (see *Define Outputs* above). If you forget how to use this button, click on any cell without a formula and click the button. This explanation will appear:

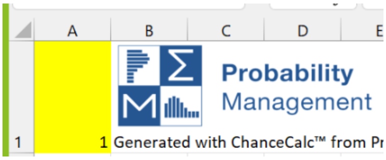


6 Displaying Individual Trials and Metadata

When you open a library a menu appears on either the first sheet of you model (or optionally a separate sheet if specified under settings) that lets you change the trial number or view metadata if present.



When building a Monte Carlo model without opening a library, you may



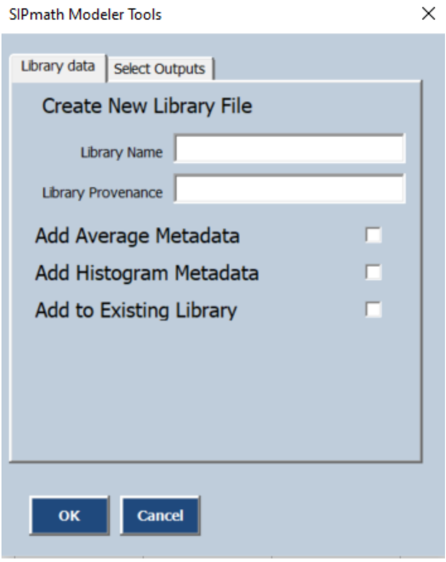
view individual trials by changing cell A1 of the PMTable sheet, which contains a cell named PM_Index.

If you do open a SIP Library, which contains metadata extensions to the distributions, you can also use those to instantiate model parameters. Keep in mind that the choice is only for your inputs. For example, if you choose “Average”, you’ll see the output values based on average inputs, NOT the average of the output values.

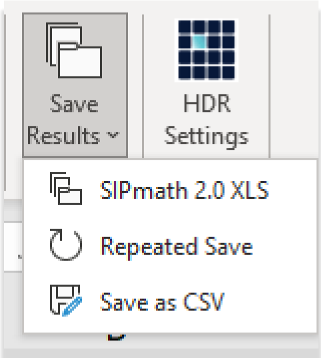
If your spreadsheet model is non-linear, you will find an interesting difference between the *output* given *average* inputs and the *average output given input distributions*. This is due to Jensen’s Inequality, commonly called the Flaw of Averages.

7 SAVE RESULTS

The *Save Results* tool allows the modeler to export the model's data to an Excel SIP library workbook. Selecting *Save Results* brings up a dialog for specifying a worksheet name and selecting the SIPs to be exported. Use *Save As* to write the SIP Library file compatible with the format expected by the Library Input tool.

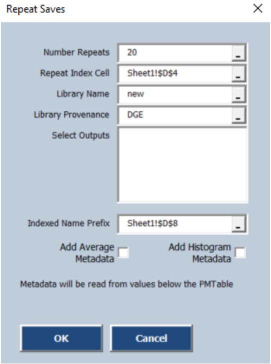


7.1 REPEATED SAVE



Pressing the bottom half of the Save Results button opens a 1-item menu giving access to the Repeated Save function.

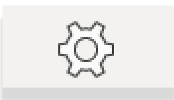
This feature allows you to run several simulations while changing the value of one simulation parameter (for example, the mean of an input distribution) and save the results in a SIP Library.



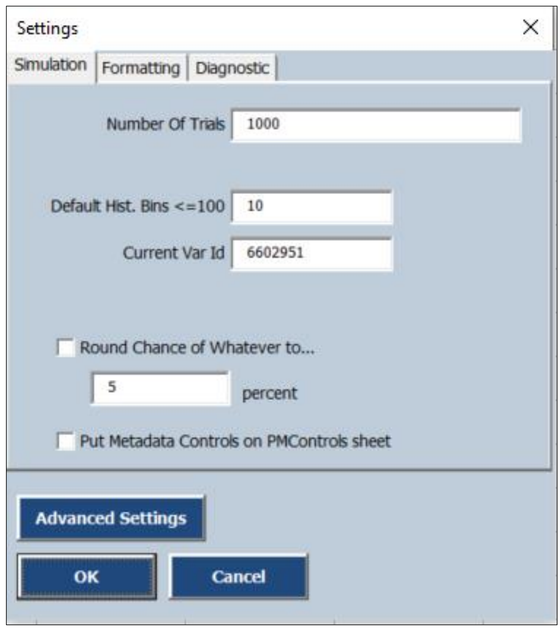
In this example, the simulation will be repeated 20 times. **Chance**Calc will put the values 1 through 20 successively into the Repeat Index Cell, D4. The Indexed Name is a cell with a text value that depends on the Repeat

Index Cell value. The results will be saved as a set of SIPs prefixed with the Indexed Name value, followed by the output name.

8 SETTINGS

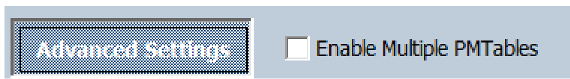


The *Settings* button is on the ChanceCalc ribbon. It brings up a dialog for setting the tool's default values and behaviors. The Number of Trials is initially set to 1,000. We can either increase or decrease that number depending on our needs. The Mode is automatically set to Generate but can be changed to SIPmath Library Mode as described in section 2. The Default Hist. Bins is 10.

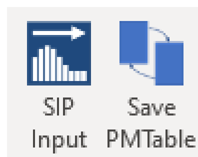


Specifying more bins provides greater definition but might also result in a more erratic looking histogram. Current Trial Index is the current value of the variable PM_Index which is defined on the PMTable worksheet. The Current Variable ID refers to the value when using the HDR random variable. This will change as random variables are defined. The Input Sparkline Color is the color of the histogram bars for Input SIP sparklines. The Output Sparkline Color is the color of the bars for Output cells. The Sparkline Border Color is the color of the border line drawn around sparkline cells. Hide Most **Chance**Calc ranges will hide or show most of the ranges defined by **Chance**Calc when you use the Name Manager command. Hiding these ranges can reduce clutter in the Name Manager. When Create Sparklines by Default is checked, **Chance**Calc will automatically fill in the currently active cell as the Starting Cell(s) for Sparklines in the *Library Input* and *Define Output* dialogs. When the field is unchecked, no cell will be filled in. (No sparkline will be created if the starting cell for sparklines is blank.)

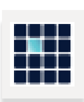
If Show SPT random cell suggestion is checked, then **Chance**Calc will show a message suggesting a good way to define random inputs to the SPT Metalog variables when you use the *SPT Metalog* command.

The Advanced Settings 

button enables the Advanced Initialize checkbox. Checking this box will activate the Save PMTable feature; the button will appear next to the Initialize button on the main **Chance**Calc toolbar.

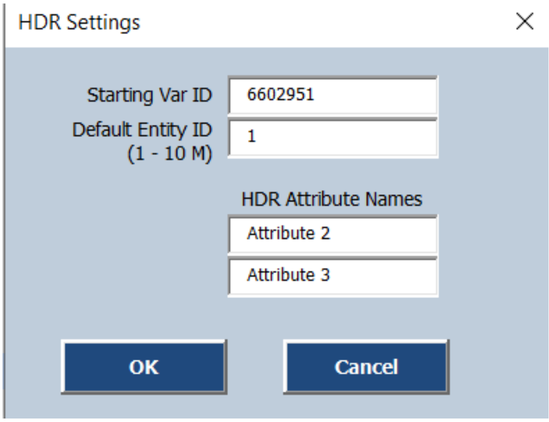


9 HDR Settings



The *HDR Settings* tool brings up a dialog in which you can specify default input values and attribute names for the HDR random number generator.

The Default Entity ID value will be used as the Entity ID parameter for all HDR random numbers generated by **Chance**Calc. It is used (for example) to distinguish departments within an organization, helping to ensure two departments won't accidentally use the same set of pseudo-random numbers.



The HDR Attribute Names are displayed in the *Generate Input* tool dialogs when you are using an HDR random number. These names are purely symbolic and have no effect on the HDR numbers. They allow you to use names that make sense when generating HDR random numbers for your application.